

Final Report

Within this report, we try to explain our methodology for solving the 1D Burgers equation with two methods one of which consolidates eventually. In fact, we have reached the conclusion that there exists no pure HSE framework for dealing with the Burgers equation, as according to the <https://arxiv.org/html/2403.00596v2>, the fluid flow required for describing this equation is the full Navier-Stokes flow, which accounts for the imaginary diffusion, behind the viscosity coefficient. This flow is much more advanced than Schrodinger's flow, which was the basis upon which the idea of Hydrodynamic Schrodinger's equation was built. According to eq.C6 in the mentioned paper, the framework required is the Schrodinger-Pauli equation, which its Hamiltonian has extra terms on top of the Schrodinger's flow. In addition, Burgers equation necessitates compressibility for its underlying fluid in the most general sense; however, this seems to be somewhat of an exception for $D=1$, where one can retrieve the VBE via removing pressure from the Incompressible SF. That caused us to consider the NSF in the SPE as the sum of two Hamiltonians: 1. The Incompressible SF (after removing pressure and the other one, capable of being approximated via a QTN framework. Consequently, there is a great difficulty embedded in capturing the essence of even 1D BE via the hybrid HSE-QTN framework.

That is the reason behind us focusing on the pure QTN framework. We construct the tensor as an array of its components and then design it in a $(2,2,2,\dots)$ form representative of the spirit of the parameters in the definition of the tensor being either 0 or 1. Then we isolate each parts of the NDAarray into a different segment where SVD is applied, and all these details are taken care of under the hood of the fascinating library TorchTT (<https://github.com/ion-g-ion/torchTT>). We performed comparison analysis of the QTT framework with the classical FDM, where we noticed much lower runtime and comparable memory usage with the FDM. This has been consistent through different number of grids.